

Bios and post troubleshooting theory

- 1) List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

BIOS/POST Error Message	Usually Points To
CMOS Checksum Error – Defaults Loaded	CMOS battery is weak/dead or BIOS settings have been reset.
Keyboard Error or No Keyboard Present	Keyboard is disconnected, faulty, or USB/PS2 port issue.
CPU Fan Error	CPU cooling fan is not connected, not spinning, or has failed.
Memory Test Failed	RAM is faulty, loose, or incompatible.
No Boot Device Found	Hard drive/SSD is not detected or there is no bootable operating system.

- 2) Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

❖ **Meaning:**

- For an AMI BIOS, 3 short beeps usually indicate a RAM (Memory) failure. The system cannot detect or properly use the installed memory, so it stops the boot process and nothing appears on the monitor.

❖ **Troubleshooting Steps:**

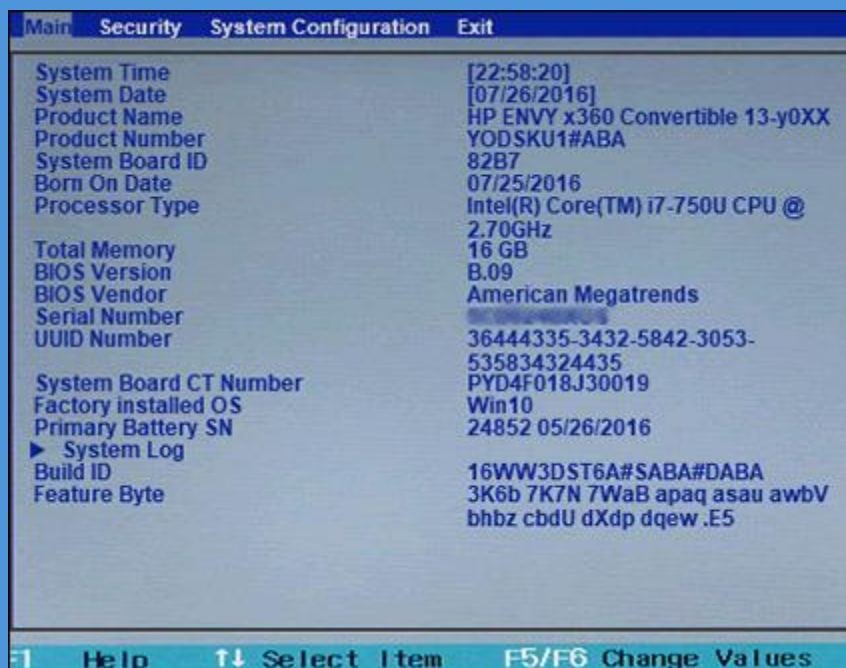
⇒ **Step 1:**

Power off the PC, unplug it, remove the RAM sticks, clean the contacts gently, and reinstall them securely. If there are multiple RAM sticks, test one at a time in different memory slots.

⇒ **Step 2:**

If reseating does not work, replace the RAM with a known working module or test the existing RAM in another compatible computer to check whether the memory is faulty.

- 3) Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



- 4) You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset..

❖ Meaning:

This error means the BIOS detected that the saved CMOS settings are invalid or corrupted. It automatically loads the default BIOS settings. The most common reason is a weak or dead CMOS battery (CR2032), although it can also happen after a BIOS reset or failed BIOS update.

❖ Resolution:

- Enter the BIOS Setup by pressing Del, F2, or the key shown during startup.
- Check and correct the date, time, and BIOS settings if needed.
- Save the settings and restart the computer.
- If the error appears again, replace the CMOS battery (CR2032) on the motherboard.
- After replacing the battery, set the correct BIOS settings again and save them.
- If the problem still continues, update the BIOS (if recommended by the motherboard manufacturer) or inspect the motherboard for faults.

